

GED Math Practice Test 18

PART A — QUESTIONS

GED-MATH18-001

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A supply room had 4 shelves with 18 manuals on each shelf and 3 boxes with 9 manuals in each box. If 21 manuals were moved to a classroom, how many manuals remained in the supply room?

- A) 72
- B) 78
- C) 99
- D) 120

GED-MATH18-002

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

What is the value of $2x^2 - 7x + 5$ when $x = 6$?

- A) 25
- B) 35
- C) 47
- D) 65

GED-MATH18-003

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A tank was $\frac{3}{5}$ full. A worker added $\frac{1}{4}$ of the tank's capacity and then used $\frac{1}{10}$ of the tank's capacity. What fraction of the tank was full after these changes?

- A) $\frac{1}{2}$
- B) $\frac{3}{5}$
- C) $\frac{3}{4}$
- D) $\frac{19}{20}$

GED-MATH18-004

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

Which expression is equivalent to $5(2a - 3) - 3(a + 4)$?

- A) $7a - 27$
- B) $7a - 3$

- C) $13a - 27$
- D) $13a - 3$

GED-MATH18-005

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A shipment contained 56 packages. If 14 packages were marked fragile, what percent of the packages were marked fragile?

- A) 20%
- B) 25%
- C) 28%
- D) 40%

GED-MATH18-006

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $R = (x - 5)^2 + 16$ is used to calculate a repair rating. If $R = 65$, what is one possible value of x ?

- A) 7
- B) 10
- C) 12
- D) 21

GED-MATH18-007

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A shipment includes 15 cartons. Each carton contains 12 packets, and each packet weighs 9.5 ounces. What is the total weight of all the packets in ounces?

Type your answer as an integer.

GED-MATH18-008

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $E = 1/2bh$ gives the area E of a triangle with base b and height h . To solve for h , first multiply both sides by $[D1]$, then divide by $[D2]$.

Dropdowns:

- D1) 2 | b | $1/2$
- D2) 2 | b | E

GED-MATH18-009

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A water storage tank held 850 gallons at the start of the week. During the week, 175 gallons were added and 390 gallons were used. What percent of the original 850 gallons remained after these changes?

- A) 54.1%
- B) 63.5%
- C) 74.7%
- D) 120.6%

GED-MATH18-010

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which statement describes all possible values of t if $5(2t + 1) - 9 \leq 66$?

- A) $t \leq 7$
- B) $t \geq 7$
- C) $t \leq 8$
- D) $t \geq 8$

GED-MATH18-011

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows the number of units completed by three teams.

Team: A, B, C

Hours worked: 3.2, 4.5, 5

Units completed: 128, 171, 205

Which team completed the greatest number of units per hour?

- A) Team A
- B) Team B
- C) Team C
- D) Teams A and B were tied

GED-MATH18-012

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

Which expressions are equivalent to $7(3x + 2) - 4(2x - 5)$?

- A) $13x + 34$
- B) $21x + 14 - 8x + 20$
- C) $13x - 6$
- D) $21x + 2 - 8x + 20$
- E) $3(4x + 9)$

GED-MATH18-013

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A nonprofit recorded donation methods for one day.

Donation method: Cash, Card, Online, Check

Donations: 18, 54, 42, 6

If one donation is selected at random, what is the probability that it was made online or by check?

- A) $1/4$
- B) $2/5$
- C) $1/2$
- D) $3/5$

GED-MATH18-014

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A formula for a point total is $P = 18r + 11s$, where r is the number of regular tasks and s is the number of special tasks. If $P = 605$ and $s = 19$, what is r ?

Type your answer as an integer.

GED-MATH18-015

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A loading area is made from a 30-foot by 12-foot rectangle with a triangular ramp attached. The triangle has a base of 12 feet and a height of 10 feet. A 5-foot by 5-foot square column space is not usable. What is the usable area?

- A) 360 square feet
- B) 395 square feet
- C) 420 square feet
- D) 445 square feet

GED-MATH18-016

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows part of a linear relationship.

x: -6, -2, 4, 10

y: 25, 15, 0, -15

Which equation represents the relationship?

A) $y = -2x + 13$

B) $y = -2.5x + 10$

C) $y = 2.5x + 40$

D) $y = -5x - 5$

GED-MATH18-017

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A store compares two coffee packages.

Package: Small, Large

Weight: 24 oz, 40 oz

Price: \$7.68, \$12.40

Which statements are true?

A) The small package costs \$0.32 per ounce.

B) The large package costs \$0.31 per ounce.

C) The small package has the lower unit price.

D) The large package costs \$0.32 per ounce.

E) The two packages have the same unit price.

GED-MATH18-018

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which inequality represents “twice the sum of a number and 8 is no less than the quotient of 72 and 3”?

A) $2x + 8 \geq 72/3$

B) $2(x + 8) \geq 72/3$

C) $2(x - 8) \geq 72/3$

D) $72/3 \geq 2(x + 8)$

GED-MATH18-019

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A customer paid \$235.40 for equipment after 7% sales tax was added. What was the equipment cost before tax?

- A) \$218.90
- B) \$220.00
- C) \$228.40
- D) \$251.88

GED-MATH18-020

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for x: $4(3x + 2) - 5(x - 6) = 101$.

Type your answer as an integer.

GED-MATH18-021

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A circular mat has an area of about 254.34 square inches. Using $\pi \approx 3.14$, what is the circumference of the mat?

- A) 28.26 inches
- B) 50.24 inches
- C) 56.52 inches
- D) 81.00 inches

GED-MATH18-022

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A line passes through the points $(-8, 20)$ and $(4, -10)$. What is the rate of change of the line?

- A) $-5/2$
- B) $-2/5$
- C) $2/5$
- D) $5/2$

GED-MATH18-023

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A department had a monthly supply budget of \$1,250. It spent \$318.50 on paper, \$247.25 on ink, and \$189.75 on mailing supplies. What percent of the budget remained?

- A) 28.4%
- B) 35.2%
- C) 39.6%
- D) 60.4%

GED-MATH18-024

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Function A is represented by the table.

x: -4, 0, 6

y: 17, 9, -3

Function B decreases by 9 when x increases by 3. Which statement is true?

- A) Function A decreases faster than Function B.
- B) Function B decreases faster than Function A.
- C) Both functions have the same rate of change.
- D) Function A has a positive rate of change.

GED-MATH18-025

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A safety supplier sold 16 tool kits at \$24.50 each, 22 pairs of safety glasses at \$8.75 each, and 12 helmets at \$18.40 each. What was the average revenue per item sold?

Type your answer as a decimal rounded to the nearest hundredth.

GED-MATH18-026

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A theater sold 68 tickets. Adult tickets cost \$12, and child tickets cost \$7. The total revenue was \$646. How many adult tickets were sold?

- A) 28
- B) 30
- C) 34
- D) 38

GED-MATH18-027

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

On a scale drawing, 4 centimeters represents 10 feet. A rectangular room is 8 centimeters by 6 centimeters on the drawing. What is the actual area of the room?

- A) 120 square feet
- B) 240 square feet
- C) 300 square feet
- D) 480 square feet

GED-MATH18-028

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The equation $8x - 5y = 2x + 30$ can be rewritten as $y = [D1]x + [D2]$.

Dropdowns:

D1) $6/5$ | $-6/5$ | $5/6$

D2) -6 | 6 | 30

GED-MATH18-029

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A box contains 6 red tokens, 9 black tokens, and 5 blue tokens. Two tokens are selected at random without replacement. What is the probability that both selected tokens are black?

- A) $9/20$
- B) $18/95$
- C) $81/400$
- D) $8/19$

GED-MATH18-030

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

The formula $A = P + Prt$ gives the total amount A after simple interest. Which equations correctly solve for t ?

- A) $t = (A - P)/(Pr)$
- B) $t = (A/P - 1)/r$
- C) $t = A - P/Pr$
- D) $t = Pr/(A - P)$
- E) $t = (A - P)r/P$

GED-MATH18-031

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A cleaning solution uses 1.25 grams of powder for each liter of water. How many grams of powder are needed for 18 liters of water?

- A) 14.4 grams
- B) 19.25 grams
- C) 22.5 grams
- D) 25.0 grams

GED-MATH18-032

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for k: $3(4k - 9) + 2(k + 5) = 95$.

- A) 7
- B) 8
- C) 9
- D) 10

GED-MATH18-033

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A cylindrical container has a diameter of 8 feet and a height of 12 feet. It is filled to 65% of its full volume. Using $\pi \approx 3.14$, how many cubic feet of liquid are in the container?

- A) 196.04 ft³
- B) 301.44 ft³
- C) 391.87 ft³
- D) 602.88 ft³

GED-MATH18-034

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The revenue R, in dollars, from selling x items is modeled by $R = -1.5x^2 + 45x$. What is the revenue when $x = 18$?

- A) \$243
- B) \$324
- C) \$486
- D) \$810

GED-MATH18-035

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A savings account has \$2,800 and grows by 4% each year. Which value is closest to the account balance after 3 years?

- A) \$2,912
- B) \$3,034
- C) \$3,150
- D) \$3,640

GED-MATH18-036

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A medicine concentration is 240 milligrams in 8 milliliters. How many milliliters are needed for a 660-milligram dose?

- A) 18 mL
- B) 20 mL
- C) 22 mL
- D) 24 mL

GED-MATH18-037

Section: Math

Type: drag_drop

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Match each target with the correct tile.

Tiles:

- A) $5x + 9y = 138$
- B) $x + y = 138$
- C) $y = x + 4$
- D) $x = y + 4$

Targets:

T1) Five small containers and nine large containers cost \$138 total

T2) The total number of small and large containers is 138

T3) The number of small containers is 4 more than the number of large containers

GED-MATH18-038

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Machine A seals 140 packages in 20 minutes. Machine B seals 126 packages in 18 minutes. If both machines run at these rates for 16 minutes, how many packages will they seal together?

- A) 112
- B) 196
- C) 224
- D) 266

GED-MATH18-039

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A storage platform can hold at most 2,250 pounds. Equipment already on the platform weighs 380 pounds, and each box weighs 95 pounds. What is the greatest number of boxes that can be added?

A) 18

B) 19

C) 20

D) 21

GED-MATH18-040

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A fish population of 6,800 decreases by 5% each quarter. Which value is closest to the population after 2 quarters?

A) 5,780

B) 6,137

C) 6,460

D) 7,497

GED-MATH18-041

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A maintenance team bought materials for a repair.

Material: Filters, Gauges, Labels

Quantity: 14, 9, 30

Unit price: \$6.20, \$13.50, \$0.45

Which statements are true?

A) The filters cost \$86.80 in total.

B) The gauges cost \$121.50 in total.

C) The labels cost more in total than the filters.

D) The total cost was \$222.80.

E) The average cost per material type was exactly \$75.00.

GED-MATH18-042

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

For the function $f(x) = 0.6x^2 - 4x + 9$, what is $f(5)$?

- A) 2
- B) 4
- C) 6
- D) 9

GED-MATH18-043

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows a relationship.

x: -2, -1, 0, 1

y: 12, 6, 2, 0

Which equation represents the relationship?

- A) $y = x^2 - 3x + 2$
- B) $y = -4x + 2$
- C) $y = x^2 + 2$
- D) $y = 3x + 2$

GED-MATH18-044

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which expression represents 10 less than the quotient of 64 and the difference of a number x and 4?

- A) $64/(x - 4) - 10$
- B) $64/x - 4 - 10$
- C) $10 - 64/(x - 4)$
- D) $(64/x - 4)/10$

GED-MATH18-045

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $S = 2\pi rh + 2\pi r^2$ gives the surface area S of a cylinder. Using $\pi \approx 3.14$, if $S = 376.8$ and $r = 5$, what is h ?

Type your answer as an integer.

GED-MATH18-046

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A rectangular hall is 44 feet by 32 feet. A semicircular stage with a diameter of 14 feet is attached inside one end of the hall. Using $\pi \approx 3.14$, what area of the hall remains outside the semicircular stage?

- A) 76.93 square feet
- B) 1,254.14 square feet
- C) 1,331.07 square feet
- D) 1,408.00 square feet

PART B — ANSWER KEY + EXPLANATIONS

GED-MATH18-001 — Correct: B — Correct Answer: 78 — Explanation: The room had $4(18) + 3(9) = 99$ manuals. After 21 were moved, $99 - 21 = 78$ remained.

GED-MATH18-002 — Correct: B — Correct Answer: 35 — Explanation: Substitute $x = 6$: $2(6^2) - 7(6) + 5 = 72 - 42 + 5 = 35$.

GED-MATH18-003 — Correct: C — Correct Answer: $\frac{3}{4}$ — Explanation: $\frac{3}{5} + \frac{1}{4} - \frac{1}{10} = \frac{12}{20} + \frac{5}{20} - \frac{2}{20} = \frac{15}{20} = \frac{3}{4}$.

GED-MATH18-004 — Correct: A — Correct Answer: $7a - 27$ — Explanation: $5(2a - 3) - 3(a + 4) = 10a - 15 - 3a - 12 = 7a - 27$.

GED-MATH18-005 — Correct: B — Correct Answer: 25% — Explanation: $\frac{14}{56} = 0.25$, so 25% of the packages were marked fragile.

GED-MATH18-006 — Correct: C — Correct Answer: 12 — Explanation: $65 = (x - 5)^2 + 16$, so $49 = (x - 5)^2$. One solution is $x - 5 = 7$, giving $x = 12$.

GED-MATH18-007 — Correct: 1710 — Tolerance: exact — Correct Answer: 1710 — Explanation: Total weight is $15 \times 12 \times 9.5 = 1,710$ ounces.

GED-MATH18-008 — Correct: $D1=2$; $D2=b$ — Correct Answer: $D1=2$; $D2=b$ — Explanation: From $E = \frac{1}{2}bh$, multiply both sides by 2 to get $2E = bh$, then divide by b .

GED-MATH18-009 — Correct: C — Correct Answer: 74.7% — Explanation: The tank had $850 + 175 - 390 = 635$ gallons after the changes. Then $635 \div 850 \approx 0.747$, or 74.7%.

GED-MATH18-010 — Correct: A — Correct Answer: $t \leq 7$ — Explanation: $5(2t + 1) - 9 \leq 66$ simplifies to $10t - 4 \leq 66$, so $10t \leq 70$ and $t \leq 7$.

GED-MATH18-011 — Correct: C — Correct Answer: Team C — Explanation: Team A: $128 \div 3.2 = 40$ units per hour. Team B: $171 \div 4.5 = 38$. Team C: $205 \div 5 = 41$.

GED-MATH18-012 — Correct: A,B — Correct Answer: $13x + 34 \mid 21x + 14 - 8x + 20$ — Explanation: $7(3x + 2) - 4(2x - 5) = 21x + 14 - 8x + 20 = 13x + 34$.

GED-MATH18-013 — Correct: B — Correct Answer: $\frac{2}{5}$ — Explanation: There were 120 donations total. Online or check donations totaled $42 + 6 = 48$. The probability is $48/120 = \frac{2}{5}$.

GED-MATH18-014 — Correct: 22 — Tolerance: exact — Correct Answer: 22 — Explanation: Substitute $s = 19$: $605 = 18r + 11(19)$, so $605 = 18r + 209$. Then $396 = 18r$ and $r = 22$.

GED-MATH18-015 — Correct: B — Correct Answer: 395 square feet — Explanation: Rectangle area is $30 \times 12 = 360$. Triangle area is $\frac{1}{2}(12)(10) = 60$. Subtract the unusable column space: $360 + 60 - 25 = 395$.

GED-MATH18-016 — Correct: B — Correct Answer: $y = -2.5x + 10$ — Explanation: From $x = -6$ to $x = -2$, y decreases by 10 while x increases by 4, so the rate is -2.5 . Substituting $x = -2$ and $y = 15$ gives $y = -2.5x + 10$.

GED-MATH18-017 — Correct: A,B — Correct Answer: The small package costs \$0.32 per ounce. | The large package costs \$0.31 per ounce. — Explanation: Small: $7.68 \div 24 = 0.32$. Large: $12.40 \div 40 = 0.31$.

GED-MATH18-018 — Correct: B — Correct Answer: $2(x + 8) \geq 72/3$ — Explanation: The sum of a number and 8 is $x + 8$. Twice that sum is $2(x + 8)$, and “no less than” means $\geq$.

GED-MATH18-019 — Correct: B — Correct Answer: \$220.00 — Explanation: The amount paid is 107% of the cost before tax. So the cost before tax was $235.40 \div 1.07 = 220.00$.

GED-MATH18-020 — Correct: 9 — Tolerance: exact — Correct Answer: 9 — Explanation: $4(3x + 2) - 5(x - 6) = 12x + 8 - 5x + 30 = 7x + 38$. Then $7x + 38 = 101$, so $x = 9$.

GED-MATH18-021 — Correct: C — Correct Answer: 56.52 inches — Explanation: Area = πr^2 , so $254.34 = 3.14r^2$. Then $r^2 = 81$ and $r = 9$. Circumference = $2(3.14)(9) = 56.52$ inches.

GED-MATH18-022 — Correct: A — Correct Answer: $-5/2$ — Explanation: Rate of change = $(-10 - 20)/(4 - (-8)) = -30/12 = -5/2$.

GED-MATH18-023 — Correct: C — Correct Answer: 39.6% — Explanation: Total spent is $318.50 + 247.25 + 189.75 = 755.50$. Remaining amount is $1,250 - 755.50 = 494.50$. Then $494.50 \div 1,250 = 0.3956$, or 39.6%.

GED-MATH18-024 — Correct: B — Correct Answer: Function B decreases faster than Function A. — Explanation: Function A has rate $(9 - 17)/(0 - (-4)) = -8/4 = -2$. Function B has rate $-9/3 = -3$, so Function B decreases faster.

GED-MATH18-025 — Correct: 16.11 — Tolerance: exact — Correct Answer: 16.11 — Explanation: Total revenue is $16(24.50) + 22(8.75) + 12(18.40) = 392.00 + 192.50 + 220.80 = 805.30$. There were 50 items sold, so $805.30 \div 50 = 16.106$, which rounds to 16.11.

GED-MATH18-026 — Correct: C — Correct Answer: 34 — Explanation: Let a be adult tickets. Then $12a + 7(68 - a) = 646$. This gives $12a + 476 - 7a = 646$, so $5a = 170$ and $a = 34$.

GED-MATH18-027 — Correct: C — Correct Answer: 300 square feet — Explanation: The scale is $10 \div 4 = 2.5$ feet per centimeter. The actual dimensions are $8(2.5) = 20$ feet and $6(2.5) = 15$ feet. The area is $20 \times 15 = 300$ square feet.

GED-MATH18-028 — Correct: $D1=6/5$; $D2=-6$ — Correct Answer: $D1=6/5$; $D2=-6$ — Explanation: $8x - 5y = 2x + 30$ gives $-5y = -6x + 30$, so $y = 6/5x - 6$.

GED-MATH18-029 — Correct: B — Correct Answer: 18/95 — Explanation: There are 20 tokens total. Without replacement, $P(\text{two black}) = 9/20 \times 8/19 = 18/95$.

GED-MATH18-030 — Correct: A,B — Correct Answer: $t = (A - P)/(Pr) \mid t = (A/P - 1)/r$ — Explanation: From $A = P + Prt$, subtract P and divide by Pr . Dividing by P first gives $A/P = 1 + rt$, so $t = (A/P - 1)/r$.

GED-MATH18-031 — Correct: C — Correct Answer: 22.5 grams — Explanation: Powder needed is $1.25 \times 18 = 22.5$ grams.

GED-MATH18-032 — Correct: B — Correct Answer: 8 — Explanation: $3(4k - 9) + 2(k + 5) = 12k - 27 + 2k + 10 = 14k - 17$. Then $14k - 17 = 95$, so $k = 8$.

GED-MATH18-033 — Correct: C — Correct Answer: 391.87 ft³ — Explanation: The radius is 4 feet. Full volume is $3.14(4^2)(12) = 602.88$ cubic feet. Sixty-five percent is $0.65 \times 602.88 = 391.872$, which rounds to 391.87.

GED-MATH18-034 — Correct: B — Correct Answer: \$324 — Explanation: $R = -1.5(18^2) + 45(18) = -486 + 810 = 324$.

GED-MATH18-035 — Correct: C — Correct Answer: \$3,150 — Explanation: After 3 years, the balance is $2,800(1.04)^3 \approx 3,149.62$, which is closest to \$3,150.

GED-MATH18-036 — Correct: C — Correct Answer: 22 mL — Explanation: The concentration is $240 \div 8 = 30$ mg/mL. For 660 mg, the dose volume is $660 \div 30 = 22$ mL.

GED-MATH18-037 — Correct: T1=A; T2=B; T3=D — Correct Answer: $T1=5x + 9y = 138$; $T2=x + y = 138$; $T3=x = y + 4$ — Explanation: Each tile matches the relationship described in the target.

GED-MATH18-038 — Correct: C — Correct Answer: 224 — Explanation: Machine A seals $140 \div 20 = 7$ packages per minute. Machine B seals $126 \div 18 = 7$ packages per minute. Together they seal 14 packages per minute, so in 16 minutes they seal 224 packages.

GED-MATH18-039 — Correct: B — Correct Answer: 19 — Explanation: The weight must satisfy $380 + 95b \leq 2,250$. Then $95b \leq 1,870$, so $b \leq 19.68$. The greatest whole number of boxes is 19.

GED-MATH18-040 — Correct: B — Correct Answer: 6,137 — Explanation: After 2 quarters, the population is $6,800(0.95)^2 = 6,137$.

GED-MATH18-041 — Correct: A,B — Correct Answer: The filters cost \$86.80 in total. | The gauges cost \$121.50 in total. — Explanation: Filters cost $14(6.20) = \$86.80$. Gauges cost $9(13.50) = \$121.50$. Labels cost $30(0.45) = \$13.50$. The total is \$221.80, not \$222.80.

GED-MATH18-042 — Correct: B — Correct Answer: 4 — Explanation: $f(5) = 0.6(5^2) - 4(5) + 9 = 15 - 20 + 9 = 4$.

GED-MATH18-043 — Correct: A — Correct Answer: $y = x^2 - 3x + 2$ — Explanation: Substituting $x = -2, -1, 0$, and 1 gives 12, 6, 2, and 0.

GED-MATH18-044 — Correct: A — Correct Answer: $64/(x - 4) - 10$ — Explanation: The difference of x and 4 is $x - 4$. The quotient of 64 and that difference is $64/(x - 4)$, and 10 less than that is $64/(x - 4) - 10$.

GED-MATH18-045 — Correct: 7 — Tolerance: exact — Correct Answer: 7 — Explanation: $376.8 = 2(3.14)(5)h + 2(3.14)(5^2)$. This gives $376.8 = 31.4h + 157$, so $219.8 = 31.4h$ and $h = 7$.

GED-MATH18-046 — Correct: C — Correct Answer: 1,331.07 square feet — Explanation: Hall area is $44 \times 32 = 1,408$ square feet. The semicircle has radius 7 feet, so its area is $1/2(3.14)(7^2) = 76.93$ square feet. The remaining area is $1,408 - 76.93 = 1,331.07$ square feet.